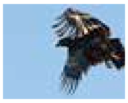

Eagles get huge roaming bill
Migrating eagles with tracking beacons that send texts reportedly accrued huge roaming charges. **https://dgit.in/nov19-65**

Aston Martin 2-wheeler
Aston is launching its first motorcycle with help from another storied British firm – Brough Superior. **https://dgit.in/nov19-66**

AMD Ryzen 5 3400G

Who needs a discrete GPU?

Price
₹12,850

It's been a couple of months since the new Zen2-based AMD 3rd Gen Ryzen processors have launched and AMD has been having a gala time given all the worldwide sales reports we've been seeing in these past few months. One of the fastest moving SKUs in any processor family are the ones which cater to the budget segment and have an Integrated Graphics Processor. The AMD Ryzen 5 3400G is one such processor. Even though it's part of the 3000 series, it's actually a Zen+ 2nd Gen processor. Blame AMD marketing for the odd naming scheme.

77

PERFORMANCE..........63
VALUE FOR MONEY....90

The AMD Ryzen 5 3400G is a 4-core/8-thread APU with a base clock of 3.7 GHz and a boost clock of 4.2 GHz. On the graphics front it has a RX Vega 11 IGP clocked at 1400 MHz which has a core configuration of 704 stream processors, 44 texture mapping units and 16 render output units. The entire thing fits within a 65 W envelope. That being said, under synthetic stress these things do consume about 115 W. While many of the initial 3rd Gen Ryzen processors did not hit their advertised frequencies with the older BIOS, that's now been fixed. Any BIOS based off the AGESA 1.0.0.3ABBA should ensure your CPU hits the mentioned specs.

What's new?

Compared to the Ryzen 5 2400G, the 3400G has slightly better clocks. The base clock is higher by 100 MHz and the boost clock is higher by 300 MHz. There's also an improvement in the overclocking headroom as a result of the improvements made in the architecture. However, the difference between Zen and Zen+ isn't much. They're both built on the TSMC 12nm process. The IGP configuration is the same but the clock has been bumped by 150 MHz. Essentially, the 3400G is very close to the 2400G. Folks who have delidded the 3400G have reported that it uses solder for the thermal interface material, so that's always a good plus.

Also, since this is a Zen+ APU, you don't get the benefits of PCIe 4.0 so clubbing a 3400G with an X570 based motherboard would be a huge waste of your money. On the other hand, if you were to pair this with a B450 based motherboard, then the value proposition is much better. You can build hunky dory PCs that can play casual games without having to install a discrete graphics card. Of course, you will have to scale everything to the lowest possible settings to get decent framerates but it's still better than nothing.

Performance

As usual, we begin our benchmarks with Cinebench R15 and the Ryzen 5 3400G scores 891 which is about 8.7 per cent faster than the 2400G in the multithreaded run. In single threaded, we ended up scoring 9.7 per cent more than the 2400G. This meagre improvement used to be the generational difference between flagships about five years ago. However, we're looking at the entry-level SKUs and in that context, this is a notable improvement.

In our Handbrake encoding tests, the Ryzen 5 3400G gained about 11 per cent on the 2400G which is, again, a noteworthy improvement on paper but when you notice that you're getting 3 extra FPS, then the difference doesn't seem to be huge. If you're planning on encoding 30-minute long videos, then it's better that you run the job overnight.

Then comes gaming. The 3400G, purely on the merit of its IGP can produce about 24-28 FPS in games such as Shadow of the Tomb Raider and Warhammer II running at 720p. Popular online multiplayer games such as CS:GO run at around 165-170 FPS and if you scale the resolution to 1440p, you can still get around 100 FPS which is pretty neat. DOTA 2 runs at a little over 60 FPS @1440p which is decent enough for it. And lastly, Fortnite can produce about 50-55 FPS @1440p.

The AMD Ryzen 5 3400G has a decent improvement in the general computing space but when it comes to gaming, it's quite the performer. You'd only need a discrete graphics card if you're playing a current gen AAA-title. Aside from that, this little APU should provide passable framerates for most casual games running at 720p and 1080p and even 1440p for some titles.

Verdict

The AMD Ryzen 5 3400G isn't a drastic improvement over the previous gen 2400G. AMD certainly could have used the new 7nm process along with the Zen2 architecture to create a killer APU but it seems we won't be seeing that for a couple of more months. Till then the 3400G is a decent improvement over the 2400G but not enough to warrant an upgrade. The 3400G does exactly what it set out to do, provide a jack of all trades in the budget segment.

–Mithun Mohandas

SPECIFICATIONS

CORES: 4 | THREADS: 8 | BASE CLOCK: 3.7 GHz | BOOST CLOCK: 4.2 GHz | PROCESS NODE: TSMC 7nm | L3 CACHE: 4 MB | L2 CACHE: 2 MB | IGP: Vega 11 | TDP: 65 W.

CONTACT

AMD | PHONE: NA | EMAIL: support.amd.com | WEBSITE: www.amd.com